GNBF5020

Practice Examination 02

Cumulative · Lectures 1–7 · 30 marks · 90 minutes · 8 questions

GNBF5020 · Introduction to Molecular Biology and Genetics

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MSc in Genomics and Bioinformatics · Lectures 1–7 · 12 Sep – 7 Nov 2026

Format: Part A 5 × MCQ (10 marks) · Part B 2 × pipeline mapping (12 marks) · Part C 1 × exception handling (8 marks) · 90 minutes

Built strictly from the course materials in this folder: the Lectures 1–5 revision pack, the Lecture 6 and Lecture 7 guides, the case study bank and the exam focus prediction

Part	What it asks for	Questions	Marks
A	Multiple choice — conceptual, single best option	5	10
B	Pipeline mapping — trace a process from input to output	2	12
C	Exception handling — find where the system breaks	1	8
	Total	**8**	**30**

[ENGLISH ONLY] ALL ANSWERS MUST BE IN ENGLISH. Dr. Wong said it three times across the lectures: *"every assignments and exam should be done in English, even this is a Chinese University of Hong Kong,"* and *"you have to answer in English, although this is a CUHK, haha, okay."* He said he gave zero marks to students who answered in Chinese. Keep gene, protein, pathway, method and drug names in English inside your English sentences.

[HOW TO ANSWER] Show the chain, not the conclusion. Every short answer uses four moves: **1.** the observation, with the number if a number was supplied; **2.** the mechanism that produces it — this is where the marks are; **3.** the clinical or experimental meaning; **4.** the action you would take, or the limit you cannot conclude past. A conclusion with no chain earns partial credit; a chain with a conclusion earns full credit.

[NO ANSWER KEY] This paper carries no answer key. The key and a full post-mortem — for every question you get wrong, a translation of the biological error into computational or data-flow terms — are returned once your answers are submitted. Sit the paper closed-book first.

[WHAT THIS PAPER IS] Deliberately beyond the Midterm Drill Pack. The five question types Dr. Wong named out loud — the Kaplan–Meier curve, the contractile ring, NHEJ→SCID, the mitochondrial pedigree and the lac operon — stay in the Drill Pack and the Exam Focus Prediction, where they belong. This paper sits past them: every question is built on mechanism and inference, and the numbers are given to you so that the reasoning, not the recall, is what is being marked.

Instructions

1. Answer in English. Dr. Wong said three times across the lectures that every assignment and exam is answered in English, and that he gave zero marks for answers written in Chinese. Keep gene, protein, pathway, method and drug names in English inside your English sentences.

2. Show the chain, not the conclusion. For every short answer, use four moves:

No.	Move	What it means
1	**Observation**	What the data shows, in one sentence, with the number if a number was supplied
2	**Mechanism**	The pathway or process that produces the observation — *this is where the marks are*
3	**Interpretation**	What it means for the patient, the assay or the experiment
4	**Action or limit**	What you would do next, or what you cannot conclude

3. State what the method cannot do. Every technique has a limit; naming it proves you know the method rather than the slogan.

4. Use the data given. Do not substitute numbers you remember from elsewhere. If a judgement requires an assumption, state the assumption explicitly.

5. Think in systems terms. The course's mechanisms are the systems you already know:

- a metabolic or signalling pathway is a **directed acyclic graph** — find the first node whose output is never consumed downstream, and that is your failure point;
- an enzyme is a **function** with a substrate and a product — ask what it consumes and what it emits;
- a regulatory state is a **logic gate** — ask how many inputs it needs, and whether the gate is open, closed, latched or amplified by feedback;
- a **feedback loop** has a sign — decide whether it stabilises the system or runs away from it.

6. Answer the question that is asked. If the verb is *explain*, naming a mechanism is mandatory; describing the data is only partial credit.

PART A — Multiple choice

(5 questions × 2 marks = 10 marks)

Five conceptual questions. Single best option only — no marks for selecting more than one letter. Each option is there to fail for a specific, identifiable reason: read them as claims, not as labels.

Single best option. Circle one letter per question. No marks for multiple selections.

Q1. The biomarker, the assay and the trial

A sponsor is preparing a phase III trial of a new HER2-directed antibody in advanced breast cancer. Two assays are on the table: a tumour-based **HER2 IHC** assay, and a **plasma ctDNA** assay that reads HER2 copy-number gain from a blood sample. The protocol will use one of them to decide which patients are enrolled.

Which statement about the biomarker, the companion diagnostic and their development is correct?

(2 marks)

A.	The biomarker is HER2 overexpression on the tumour cell surface — the biological entity — and the companion diagnostic is the *assay* that measures it, whichever assay the sponsor chooses and validates. The CDx is not optional: it is developed **alongside** the drug, and it has to be ready before recruitment opens, because the trial screens every incoming patient with it. A biomarker without a validated CDx is a biological observation that nobody can act on.
B.	The biomarker and the companion diagnostic are the same thing, so once HER2 IHC has been chosen as the platform the distinction is settled, and the ctDNA assay is simply a cheaper version of the same test.
C.	The companion diagnostic can be filed after launch, because the antibody works in HER2-positive disease whatever laboratory measures HER2, and only the safety database needs to know how the patients were selected.
D.	HER2 status is prognostic, so HER2-negative patients can be spared the antibody and HER2-positive patients should receive it, and this holds without any treatment comparison.
E.	The plasma ctDNA assay should replace IHC outright, because a blood draw is less invasive and can be repeated, and because one plasma assay can also report the T790M resistance mutation that is monitored after a patient progresses.

Q2. One score, two claims

A breast cancer cohort has been genotyped with a **21-gene RT-qPCR** panel — **16** cancer-related genes plus **5** reference genes — which produces a **Recurrence Score (RS) from 0 to 100**. Two published results are quoted.

Prognosis. In a multivariate Cox analysis the RS was the only clinical factor whose confidence interval excluded 1: **HR 2.81 (95 % CI 1.70–4.64), P < 0.001, per 50-point increment** of RS. The algorithm was derived in cohorts that included the tamoxifen arm of **NSABP B-20**, and it was then prospectively validated on archival **NSABP B-14** blocks, in which RT-PCR succeeded in **668 of 675** blocks. The score also stratifies 10-year distant recurrence:

RS band	10-year distant recurrence
< 18 (low)	**6.8 % (4.0–9.6)**
18 to < 31 (intermediate)	**14.3 % (8.3–20.3)**
31 to 100 (high)	**30.5 % (23.6–37.4)**

Prediction. In **NSABP B-20**, **651** node-negative, ER+ patients had been randomised to **tamoxifen alone (227)** or **tamoxifen plus chemotherapy (424)**. The biomarker-by-treatment interaction test gave **P = 0.038**; the benefit of chemotherapy was confined to the **RS ≥ 31** group, and in the low-RS group the odds ratio for chemotherapy exceeded 1.

For comparison, the **70-gene** MammaPrint signature is a **DNA-microarray** assay reporting high risk or low risk, and it was validated in an independent cohort of **295** patients.

Which statement about these results is licensed? (2 marks)

A.	The RS is predictive, because the analysis gives a hazard ratio above 1 with a confidence interval that excludes 1.
B.	The RS is prognostic only, because the 6.8 %, 14.3 % and 30.5 % recurrence figures are outcome data, and outcome data carry no treatment comparison.
C.	The RS is both. It is prognostic, because it independently predicts distant recurrence (HR 2.81 per 50-point increment), and predictive, because in the randomised trial with a treatment arm the score separated the patients who gained from chemotherapy from those who did not. The treatment arm is what licenses the second claim; the prognostic claim does not need one.
D.	Both assays are ready to decide who receives chemotherapy, because the 70-gene signature was validated in 295 independent patients and the 21-gene algorithm in 668 of 675 archival blocks.
E.	The RS is a predictive marker for chemotherapy, because a high score predicts 30.5 % distant recurrence and a low score predicts 6.8 %.

Q3. An assay that measures the right thing in the wrong range

A new plasma ELISA for a candidate biomarker measures reliably from **10 to 100 ng/mL**. In patients with the disease of interest the biomarker circulates at **100 to 1,000 ng/mL**. The assay passes every other analytical check in the laboratory's validation file.

Which statement is correct? (2 marks)

A.	The failed parameter is **specificity**: the antibody is cross-reacting with homologous proteins, so the assay must be rebuilt as a monoclonal sandwich before it can be used — and once specificity is fixed, the test is licensed for clinical use.
B.	The failed parameter is the **dynamic range**: the assay's quantification range (10–100 ng/mL) stops where the clinical range (100–1,000 ng/mL) begins, so it cannot quantify the values that decide management. The range has to be re-developed — by specimen dilution, by sample pre-treatment, or by a new calibration and capture range — so that it covers 100–1,000 ng/mL. Passing analytical validation still licenses nothing clinically: analytical validation shows that the assay measures its target correctly, while clinical validity (correlation with the clinical endpoint) and clinical utility (using the test changes management and improves outcome) are separate proofs that have to be supplied in turn.
C.	The failed parameter is the **lower limit of quantification**: the LLoQ must be lowered, because the assay cannot detect the biomarker at the low end where early disease sits.
D.	The failed parameter is **reproducibility**: a single validation run cannot support a release, so the assay has to be repeated across days, operators and reagent lots before the range is even relevant.
E.	No analytical parameter has failed: the assay works exactly as designed, the biomarker simply has a wide biological range in patients, and the test is therefore ready for clinical use once a cut-off is chosen.

Q4. Driver or passenger

A tumour has been profiled by RNA-Seq and copy-number analysis. One gene, **G1**, is **amplified**, **highly expressed**, and sits in a strongly **enriched pathway**. Three other genes, **G2**, **G3** and **G4**, are co-expressed with G1 at similar levels but show no copy-number change. The sponsor wants to make G1 the target of a new drug.

Which statement is correct? (2 marks)

A.	G1 is the driver because it is the most highly expressed of the four genes.
B.	G1 is the driver because it is amplified; the enriched pathway and the expression level are both consequences of that amplification and add nothing to the call.
C.	G1 is the driver because it sits in the enriched pathway; any gene in a deregulated pathway is therefore a driver.
D.	The driver call is licensed by the **convergence** of three independent lines — copy-number amplification **and** up-regulation **and** membership in an enriched pathway — and expression alone cannot do it, because passengers are co-expressed with drivers, so G2, G3 and G4 cannot be separated from G1 on expression alone. Neither amplification nor enrichment is decisive on its own either: a bystander region can be amplified by chance, and a passenger can sit in an enriched pathway. The call remains an inference rather than a proof, and a causal claim needs functional evidence, such as a CRISPR knock-out or knock-in with an isogenic control.

Q5. Two ways to count a rare molecule

A patient with lung cancer has progressed on a tyrosine kinase inhibitor and the tumour is suspected of carrying the **EGFR T790M** resistance mutation. The team wants to detect it in **plasma ctDNA** rather than by re-biopsy, and to read several targets at once from a small volume of blood.

Two assays are on the table:

- **Assay 1 — qPCR with a standard curve.** The unknown is read off a curve built from **10, 5, 2.5 and 1.25 ng** of mutant DNA — a two-fold serial dilution across four tubes.
- **Assay 2 — droplet digital PCR (ddPCR).** The sample is partitioned into thousands of droplets, each droplet is amplified, and the positive and negative droplets are counted. **No standard curve** is used.

Which assay is the better choice for this readout, and why? (2 marks)

A.	Assay 1, because the standard curve built from 10, 5, 2.5 and 1.25 ng of mutant DNA is what converts the unknown's Ct into an absolute concentration, and an absolute number is what a rare mutation needs in order to be detected.
B.	Assay 2, because it needs **no standard curve** at all: it partitions the sample into thousands of droplets, counts the positive and negative droplets, and converts those counts into an absolute concentration by **Poisson statistics**. It is also **multiplex** — several targets can be read in one reaction from a small volume — which is what a rare, low-abundance ctDNA readout needs.
C.	Assay 1, because a standard curve is required to quantify a rare mutation, and ddPCR cannot give an absolute concentration without one.
D.	Assay 2, but only because it is the more sensitive of the two; the droplet counts still have to be read against a standard curve before they mean anything.

E.	Either is acceptable, and the choice should be made on what the laboratory already owns and on cost, because the two methods produce the same answer once the data have been collected.

PART B — Pipeline mapping

(2 questions × 6 marks = 12 marks)

Two long-answer questions. Trace the process from input to output, stage by stage, and say at each stage what it consumes, what it emits, and what it cannot establish.

Trace a biological process from input to output, stage by stage. At each stage name the input, the transformation, the output, and — where the course states it — one thing that stage cannot establish.

Q6. From a gene list to a drug a patient can be prescribed

A tumour has been profiled by RNA-Seq, and **412** genes are differentially expressed between the tumour and matched normal tissue. The sponsor's bioinformaticians will use **g:Profiler** and **GSEA** for the enrichment step, and **Cytoscape** with the **EnrichmentMap** app to draw the pathway network. The sponsor wants a drug that a patient can actually be prescribed: the molecule has to survive toxicology and the IND, and the patients who should receive it have to be identifiable in clinic.

Answer the following.

(a) Write out the pipeline, in order, that turns this gene list into a drug a patient can be prescribed. At each stage name the technique, its input, its output, and the stage it feeds. **(2 marks)**

(b) State what the pathway enrichment analysis cannot establish on its own, and what the copy-number analysis cannot establish on its own. **(1 mark)**

(c) State what has to happen between the target and the IND dossier — name the toxicology that is required and the meeting that precedes the submission — and state what that toxicology package cannot establish. **(1 mark)**

(d) Explain why the companion diagnostic is developed **alongside** the drug rather than after it, and name the two recruitment designs and what each one does. **(1 mark)**

(e) Say where the chain breaks if the CDx is not ready before recruitment opens, and state what the treatment decision at the end of the pipeline actually licenses. **(1 mark)**

Q7. From a blood sample to a test that changes management

A team is developing a blood-based test to decide which patients with advanced solid tumours should receive a new targeted agent. The intended readout is **circulating tumour DNA (ctDNA)** quantified by **droplet digital PCR**, and the regulatory route is a companion diagnostic filed alongside the drug.

Worked example — clinical validity. A candidate early-HCC blood test was built as a two-miRNA classifier, **miR-15b + miR-130b**. Discovery profiled **96** HCC tumours with matched adjacent non-tumour tissue; in a **15**-patient longitudinal subset the serum levels fell after the tumour was removed; the classifier was then tested in an independent cohort of **30** healthy controls, **29** chronic hepatitis B carriers and **57** HCC patients, giving **AUC 0.981**, **sensitivity 98.25 %**, **specificity 91.53 %** and **accuracy 94.83 %**.

Answer the following.

(a) Write out the pipeline, in order, from the blood sample to a test the sponsor can put in a protocol. At each stage name the input, what is measured or established, and the output. **(2 marks)**

(b) Name the analytical parameters that have to be established, state in one line what each one catches, and say what the whole analytical package still cannot establish. **(1 mark)**

(c) State what clinical validity is, how it is established, quote the numbers from the worked example that show the miRNA classifier separates HCC from the comparison groups, and state what clinical validity still does not license. **(1 mark)**

(d) Define clinical utility, state the evidence that establishes it, and explain why an analytically perfect assay with no clinical utility changes nothing for a patient. **(1 mark)**

(e) State what the regulatory label has to say, and name the joint FDA–NIH resource used to qualify biomarkers and endpoints. **(1 mark)**

PART C — Exception handling

(1 question × 8 marks = 8 marks)

One scenario. Find the point where the system breaks, follow the consequence downstream, and say how you would prove where it broke.

One scenario. Trace where the system breaks, what happens downstream, and how you would find it.

Q8. The biomarker that worked analytically and failed clinically

A sponsor is running a randomised phase III trial of a new antibody for advanced non-small-cell lung cancer, against chemotherapy alone. The sponsor's companion diagnostic is a **plasma ctDNA** assay run centrally at screening, which calls each patient **biomarker-positive** or **biomarker-negative** against a cut-off locked in the protocol. The CDx was developed alongside the drug, and it passes every analytical parameter the laboratory checks: lower limit of quantification, specificity and cross-reactivity, dynamic range, reproducibility across days, operators and reagent lots, and matrix effect including autoantibody and anti-drug antibody interference.

Every patient who passes screening was randomised — an **all-comer design**, because the sponsor had to know whether the antibody's benefit was confined to the biomarker-positive group. The CDx result was recorded for every patient before randomisation, and the two subgroup analyses were pre-specified.

Stratum	n	HR, new antibody vs chemotherapy alone	P
Biomarker-positive	240	1.06 (95 % CI 0.80–1.41)	0.69
Biomarker-negative	180	0.58 (95 % CI 0.38–0.88)	0.008
Biomarker-by-treatment interaction			0.02

The sponsor's plan now is to launch a confirmatory trial that enrols **biomarker-positive patients only** — an enrichment design — and to file the antibody as a therapy indicated for biomarker-positive disease. The clinical team disagrees.

Answer the following.

(a) Locate the break in the pipeline and trace the downstream consequence. Say which stage has passed, which stage has failed, what has to change in the sponsor's plan for the CDx, for the trial and for the label, and what has to change in the biomarker hypothesis itself. **(3 marks)**

(b) State what the biomarker was actually shown to predict, and state what it was not shown to do. **(1 mark)**

(c) State what evidence would be needed to make a predictive claim from this biomarker, and say which rung of the validation ladder that evidence has to come from. **(2 marks)**

(d) State what an enrichment design controls for, what it cannot control for, and what would have happened to this trial if the sponsor had used one from the start. **(1 mark)**

(e) State what the team should do next, in order. **(1 mark)**

When you have finished

Submit your answers. The **answer key and a full post-mortem** — including, for every question you get wrong, a translation of the biological error into computational or data-flow terms — will be returned once your answers are in.

A note on marking. Part A is worth 10 of the 30 marks and is close to pure recognition. The 20 marks in Parts B and C are where the grade is won, and they are won by the chain, not by the conclusion. A conclusion with no chain earns partial credit; a chain with a conclusion earns full credit.

Source basis: `GNBF5020_Complete_Revision_All_Lectures.md` *(Lectures 1–5, audio + slides),* `GNBF5020_Lecture_6_Exam_Focus_and_Learning_Guide.md` *(Lecture 6),* `GNBF5020_Lecture_7_Exam_Focus_and_Learning_Guide.md` *(Lecture 7, including its Appendix C corrections),* `GNBF5020_Case_Studies.md`*. The microarray-vs-RNA-seq cover plate in the Lecture 6 guide is sourced to Rao MS et al., Frontiers in Genetics 2019;9:636 (doi:10.3389/fgene.2018.00636). Every mechanism and every quoted figure comes from those course materials; the trial data in Q8 are constructed for the scenario. Questions are original to this paper and sit beyond the Midterm Drill Pack.*